The National Heritage Fellowship was created to honor exceptional folk and traditional artists in the United States. One artist who received the fellowship is Navajo (Diné) basket weaver Mary Holiday Black. Black was chosen for her lifetime _____ the arts.

1 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) contributions to

(B) doubts about

(C) imitations of

(D) misunderstandings of

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HighestScore scoring service

The following text is from Kenneth Grahame's 1908 novel *The Wind in the Willows*. The Mole is returning home after a visit to Mr. Badger's house.

> As he hurried along, eagerly anticipating the moment when he would be at home again among the things he knew and liked, the Mole saw clearly that he was an animal of tilled field and hedge-row, linked to the ploughed furrow, the frequented pasture, the lane of evening lingerings, the cultivated garden-plot.

2 Mark for Review

As used in the text, what does the word "anticipating" most nearly mean?

(A) Describing

(B) Getting ahead of

(C) Looking forward to

(D) Instructing

@JohannesKP
HighEdScoring service

The Fly River delta is a remarkably _____ landscape: it is a constantly evolving network of channels and strips of land that change in size and shape as the river deposits new sedimentary particles where the river meets the waters of the Gulf of Papua.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) mutable

(B) habitable

(C) secluded

(D) homogeneous

@JohannesKP Highest scoring service

A microgenre is a specialized genre consisting of a comparatively small number of stylistically similar artists. The microgenre of electronic music known as hyperpop emerged in the 2010s, with American singer Dorian Electra as an early exponent. Their combination of lush synthesizer arrangements and propulsive beats with vocals electronically shifted in pitch below their natural range exemplifies the hyperpop sound. More recently, Japanese-British recording artist Rina Sawayama has contributed to the microgenre by incorporating pop melodies into hyperpop songs.

5 Mark for Review

Which choice best describes the function of the underlined sentence in the text as a whole?

(A) It anticipates an objection to the text's endorsement of hyperpop.

(B) It notes an exception to the text's description of hyperpop.

(C) It defines a term used in the text's discussion of hyperpop.

(D) It states that the text's intended audience mainly consists of hyperpop fans.

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HighEst Scoring service

Cuttlefish and toads see in three dimensions (3D) by combining two images in their brains, one from each eye. This produces a sense of depth, helping the animals judge how close or far away an object is. Researchers have investigated 3D vision in praying mantises as well. In one study, Vivek Nityananda and his team fitted mantises' faces with two different color filters, one covering each eye, much like the filters in 3D glasses once worn at movies. By observing the mantises' reaction to projected images, the team confirmed that mantises do indeed have 3D vision, but it's unlike that of other animals.

6 Mark for Review

Which choice best states the function of the underlined portion in the text as a whole?

(A) It identifies a potential problem that the researchers faced while studying the praying mantises.

(B) It offers a comparison meant to aid understanding of the praying mantis study.

(C) It emphasizes a difference between the research on praying mantis vision and research on other animals' vision.

(D) It describes an earlier use of a tool the researchers used in the praying mantis study.

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Highest scoring service

Text 1

The island of Grande Terre split from the former supercontinent Gondwana around 80 million years ago, carrying Gondwanan species from a variety of clades with it. The island was periodically submerged until 37 million years ago, but some researchers suggest that its current biota includes species from clades predating the split that took refuge on islands near Grande Terre during submergence events and then returned.

Text 2

Thomas R. Buckley et al. found that the crown age—the age of the most recent common ancestor of all species in the clade (i.e., the clade's founder)—of Grande Terre's clade of stick insects is 41.1 million years, which is among the oldest clade crown ages of species inhabiting the island today. Nearly all of Grande Terre's living species belong to clades that originated much more recently: for example, the crown age of the island's clade of Goodeniaceae plants is 2.0 million years.

7 Mark for Review

Based on the texts, the author of Text 2 would most likely agree with which statement about the "Gondwanan species" discussed in Text 1?

(A) Most of them began recolonizing Grande Terre from nearby islands around 2.0 million years ago.

(B) Clades to which they belong originated no earlier than 41.1 million years ago.

(C) Few if any of them were members of a clade that includes species currently inhabiting Grande Terre.

(D) Although most of them have living descendants on Grande Terre, the stick insects and Goodeniaceae plants do not.

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Text 1

According to a study by a conservation group representing 11 tribal nations in the Great Lakes region, the northern wild rice (*manoomin* in the Ojibwe language) will have significantly worse outcomes over the next 50 years if temperatures increase as much as some models suggest. By contrast, the white-tailed deer (*waawaashkeshi* in Ojibwe) should be able to withstand the highest predicted warming without much harm and so likely won't require the conservation efforts that the northern wild rice will.

Text 2

US government agencies involved in conservation are unfortunately not able to address every possible threat to natural resources. They must use the best information available to decide which species are most threatened and therefore most in need of conservation efforts.

8 Mark for Review

Based on the texts, both authors would most likely agree with which statement?

(A) State, federal, and tribal groups involved in natural-resource management in the Great Lakes region should immediately begin conservation programs for both the northern wild rice and white-tailed deer.

(B) Agencies involved in natural-resource management in the Great Lakes region should focus their conservation efforts more on the northern wild rice than on the white-tailed deer.

(C) A collaborative approach is necessary to keep temperatures in the Great Lakes region from increasing to the highest predicted levels.

(D) Conservation efforts focused on the northern wild rice are more likely to be successful if they incorporate state and federal agency resources with the knowledge of tribal groups in those efforts.

The following text is from Julia Alvarez's 2000 novel *In the Name of Salomé*. Salomé, a poet, is hosting guests in the front parlor of her family home, and Ramona is her sister. A salon is a social gathering for the exploration of intellectual ideas.

> It was evening when the two men got up to leave. Tía Ana had already come into the room several times to see if these guests had departed yet. The front parlor had always been her special province, as she used it for her little school. Now, every evening, it turned into Salomé's salon, as Ramona called it, and it was never in order for its transformation back to a classroom the following morning.

©2000 by Julia Alvarez

9 Mark for Review

Based on the text, what most likely motivates Tía Ana's behavior during Salomé's salon?

(A) She considers the guests to be uninteresting and is trying to convince them to leave.

(B) She is impatient to share her plans to start a new school with the guests and hopes they will support her.

(C) She is anxious for the gathering to disperse so that she can ready the space for her own needs.

(D) She is frustrated because she needs assistance elsewhere in the house, but Salomé is unavailable while entertaining the guests.

@JohannesKP

Effect of Neighboring Species on Pollinator Visits to Target Species

Neighboring species	Target species	Effect value
sticky catchfly	common cow-wheat	0.2379
leafy spurge	purple locoweed	-0.8428
prickly pear	sea heath	-0.4703
Canadian wood betony	mayapple	0.4729

Researchers Carolina Laura Morales and Anna Traveset gathered data about flowering plants growing alongside each other in various locations. In each case, the researchers identified one plant as a "target species" and a nearby plant as a "neighboring species." The researchers then calculated a value to show how the neighboring species affected pollinator visits to the target species. A negative effect value indicates that the neighboring species had a harmful effect on the target species. Based on the table, two neighboring species that had a harmful effect on target species are the _____

11 Mark for Review

Which choice most effectively uses data from the table to complete the statement?

(A) prickly pear and the Canadian wood betony.

(B) leafy spurge and the prickly pear.

(C) leafy spurge and the sticky catchfly.

(D) Canadian wood betony and the sticky catchfly.

@JohannesKP

"A Pair of Silk Stockings" is an 1897 short story written by Kate Chopin. In the story, Mrs. Sommers attends a play, which she experiences as a temporary escape from the circumstances of her daily life: _____

12 Mark for Review

Which quotation from "A Pair of Silk Stockings" most effectively illustrates the claim?

(A) "The play was over, the music ceased, the crowd filed out. It was like a dream ended."

(B) "But there were vacant seats here and there, and into one of them she was ushered, between brilliantly dressed women who had gone there to kill time and eat candy and display their gaudy attire."

(C) "It was a little later when she entered the theatre, the play had begun and the house seemed to her to be packed."

(D) "When she entered [the restaurant] her appearance created no surprise, no consternation, as she had half feared it might."

@JohannesKP

Hypothesizing that lullabies, characterized by their slow tempos, are universally calming to infants, Constance M. Bainbridge and colleagues played a lullaby sung in the Western Nahuatl language and a non-lullaby sung in the Serbian language to a group of infants. The team found that the infants' heart rates and pupil size both decreased more during the lullaby than during the non-lullaby. Since a decrease in heart rate is associated with relaxation, the team concluded that the lullaby relaxed the infants. However, noting that reduced heart rate can also be associated with increased attention, one critic argues that instead, the lullaby simply attracted the infants' attention.

13 Mark for Review

Which finding, if true, would most directly weaken the critic's claim?

(A) Parents of infants in the study preferred the Western Nahuatl lullaby over the Serbian non-lullaby.

(B) Infants in the study had never heard the Western Nahuatl lullaby before.

(C) More frequent blinking has also been found to be a reliable indication of attention.

(D) Pupil size typically increases when a stimulus captures a person's attention.

@JohannesKP

Composition and Fracture Toughness of Five HEAs

HEA identification number	Composition (%)	Fracture toughness (megapascals times the square root of crack length)
15	chromium (33.33), cobalt (33.33), nickel (33.33)	265.20
7	chromium (20), cobalt (20), iron (20), manganese (20), nickel (20)	219.00
25	aluminum (1.07), carbon (46.78), chromium (1.07), cobalt (1.07), copper (1.07), iron (1.07), nickel (1.07), tungsten (46.78)	10.41
95	aluminum (20), cobalt (20), copper (20), nickel (20), zinc (20)	4.45
51	molybdenum (25), niobium (25), tantalum (25), tungsten (25)	3.30

Composition values may not add up to 100 due to rounding.

High-entropy alloys (HEAs) have been observed to have greater fracture toughness—greater resistance to crack propagation—than conventional alloys. *It* has been proposed that fracture toughness increases with the proportion of an HEA consisting of cobalt, but data on HEAs compiled by engineer Xuesong Fan show that this is not the case; for example, _____

14 Mark for Review

Which choice most effectively uses data from the table to complete the text?

(A) cobalt constitutes the same proportion of HEA 7 as it does of HEA 95.

(B) cobalt constitutes a higher proportion of HEA 15 than it does of HEA 7.

(C) cobalt constitutes a substantial proportion of HEA 15 but does not constitute any of HEA 51.

(D) cobalt constitutes a different proportion of HEA 25 than it does of HEA 7.

The olona shrub is one of many forest plant species native to Oahu (a Hawaiian island) that are at risk of extinction. The survival of most of these species in the wild largely depends on birds eating their fruits and then dropping the seeds in different locations. Although Oahu's native fruit-eating birds have all gone extinct, the common waxbill and other fruit-eating bird species have been brought to the island and are now common there. Studies confirm that these non-native birds are spreading plant seeds on Oahu, suggesting that the birds _____

15 Mark for Review

Which choice most logically completes the text?

(A) are dropping higher numbers of native forest plant seeds around the island than native bird species did in the past.

(B) show significantly more interest in eating the fruits of native forest plants than in eating the fruits of non-native ones.

(C) may be necessary for the continued survival of vulnerable forest plant species, such as the olona shrub.

(D) may also engage in other activities that affect the ability of olona shrubs and other vulnerable forest plants to continue to spread to new areas.

@JohannesKP

The human body has three types of muscle ____ cardiac, and skeletal. The levator anguli oris is a skeletal muscle—of which the body contains more than six hundred—and it helps with raising the corners of the mouth.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) tissue and smooth,

(B) tissue; smooth,

(C) tissue smooth,

(D) tissue. Smooth

@JohannesKP
Highest scoring service

Alabama resident Benjamin S. Turner, sworn in as a member of the US House of Representatives in 1871, was one of the nearly two thousand African _____ during the decade following the Civil War.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) Americans' who won elections'

(B) Americans who won election's

(C) American's who won elections

(D) Americans who won elections

@JohannesKP

The programming languages COBOL, developed by Grace Hopper in _____ developed by Rich Hickey in 2007, are all routinely translated into executable code by tools known as compilers.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) 1959, Java developed by James Gosling in 1995; and Clojure

(B) 1959, Java; developed by James Gosling in 1995; and Clojure,

(C) 1959; Java, developed by James Gosling in 1995, and Clojure,

(D) 1959; Java, developed by James Gosling in 1995; and Clojure,

@JohannesKP

Proto-Kru is a protolanguage, or a hypothesized ancestral language, of all the Kru languages (thirty-eight languages that, because _____ descendants of Proto-Kru, can provide information about the protolanguage's structure).

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) its

(B) they're

(C) their

(D) it's

Though Middle English was widely spoken in fourteenth-century England, the English language was rarely employed in literature until poet Geoffrey Chaucer pioneered its literary use. _____ his manuscripts contain the first documented uses of over 2,000 English words—like the word "digestion" in his 1395 poem "The Squire's Tale"—which led a contemporary to dub him "the first finder of our fair language."

21 Mark for Review

Which choice completes the text with the most logical transition?

(A) However,

(B) Besides,

(C) In fact,

(D) On the contrary,

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In skateboarding, the 900—a trick in which the skateboarder spins two and a half times in midair—is so rare that every successful execution of it is a historic occasion. _____ ever since Mitchie Brusco and Tas Pappas performed their 900s (in 2011 and 2014, respectively), fans have revered them as titans of the sport.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) In conclusion,

(B) By comparison,

(C) For this reason,

(D) Regardless,

@JohannesKP Highest scoring service

Visually distinguishing the common raven (*Corvus corax*) from the Chihuahuan raven (*Corvus cryptoleucus*) can confound even seasoned bird watchers. _____ the two species share similarities that—at times—make the birds appear virtually identical. Size, though, remains a differentiating feature: the common raven tends to be larger.

23 Mark for Review

Which choice completes the text with the most logical transition?

(A) Instead,

(B) Moreover,

(C) Indeed,

(D) Thus,

@JohannesKP
Highest scoring service

While researching a topic, a student has taken the following notes:

- Elizabeth Catlett (1915–2012) was a celebrated African American artist.
- She is best known for creating sculptures and prints that explore the Black experience.
- *Mahalia* is a 2002 sculpture by Catlett.
- *Paulina* is a 2009 print by Catlett.

24 Mark for Review

The student wants to provide an example of one of Catlett's sculptures. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The sculpture *Mahalia* was created by celebrated artist Elizabeth Catlett in 2002.

(B) Elizabeth Catlett, a celebrated artist, was born in 1915.

(C) The print *Paulina* was created by celebrated artist Elizabeth Catlett in 2009.

(D) Artist Elizabeth Catlett is best known for creating sculptures and prints that explore the Black experience.

While researching a topic, a student has taken the following notes:

- Lighthouses send out crucial light signals to help ships and other watercraft navigate at night.
- Before automation, lighthouses were run by lighthouse keepers.
- Kate Walker was the lighthouse keeper at Robbins Reef Light in New York.
- She held this position from 1894 to 1919.
- Georgiana C. Brumfield was the lighthouse keeper at Turkey Point Light in Maryland.
- She held this position from 1895 to 1919.

25 Mark for Review

The student wants to emphasize a similarity between Kate Walker and Georgiana C. Brumfield. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Lighthouse keepers during the late nineteenth and early twentieth centuries were crucial to ensuring safe navigation for watercraft.

(B) From 1894 to 1919, the nighttime waters of New York were more navigable thanks to Kate Walker.

(C) Kate Walker and Georgiana C. Brumfield were both lighthouse keepers during the late nineteenth and early twentieth centuries.

(D) Kate Walker and Georgiana C. Brumfield spent their careers as lighthouse keepers in different lighthouses.

@JohannesKP

- The University of Oslo in Oslo, Norway, is home to a Foucault pendulum.
- The pendulum consists of a weighted ball that swings at the end of a roughly 14-meter-long cable.
- Like all Foucault pendulums, it dangles from a fixed point that ensures the swing path of the pendulum doesn't change.
- To an observer, the swing path of a Foucault pendulum appears to change over time because Earth rotates beneath it.
- Foucault pendulums are used as a simple way to provide evidence of Earth's rotation.

26 Mark for Review

Which choice most effectively uses information from the given sentences to specify the length of the Foucault pendulum's cable?

(A) With a swing path that appears to change over time, a Foucault pendulum provides evidence of Earth's rotation.

(B) The Foucault pendulum at the University of Oslo consists of a weighted ball and a cable.

(C) Although the swing path of a Foucault pendulum doesn't actually change, it appears to change due to Earth rotating beneath the pendulum.

(D) The Foucault pendulum at the University of Oslo in Oslo, Norway, includes a cable that is roughly 14 meters long.

While researching a topic, a student has taken the following notes:

- Mauritania is a country in North Africa.
- A high percentage of Mauritania's population (39.9 percent) is under fifteen years old.
- It has the thirty-fourth-largest under-fifteen population in the world.
- Roughly 40 percent of Africa's population is under fifteen years old—the highest of any continent.
- According to the United Nations (UN), Africa's "high number of young people is an opportunity for the continent's growth—but only if these new generations are fully empowered to realise their best potential."

27 Mark for Review

The student wants to emphasize the global rank of Mauritania's youth population. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) With 39.9 percent of its population under fifteen years of age, Mauritania has the thirty-fourth-largest population for that age range in the world.

(B) Africa's high population of young people is due in part to the high percentage of young people in Mauritania.

(C) Making up roughly 40 percent of the continent's total population, Africa's under-fifteen population offers "an opportunity for the continent's growth," according to the UN.

(D) "Only if these new generations are fully empowered to realise their best potential," says the UN, will Africa's high percentage of young people lead to the continent's growth.

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Highest scoring service

Jessica Watson, who was the youngest person to sail nonstop and unassisted around the world, undoubtedly accomplished much, but to gain a lasting place in our historical memory, there is little that can _____ being the first to do something. For example, people will always remember that Jeanne Baré was the first woman to circumnavigate the world.

1 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) overreach by

(B) prevail over

(C) fluctuate with

(D) constrain within

@JohannesKP
Highest scoring service

In the mid-1980s, the price of vintage and even new baseball cards rose dramatically, which had the counterintuitive effect of _____ demand: buyers who hadn't previously wanted to purchase baseball cards thronged the market, believing prices would continue to rise and the cards could be resold later at a profit.

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) capitalizing

(B) exploiting

(C) eliciting

(D) satisfying

@JohannesKP
Highest scoring service

The fact that publications by University of California-Berkeley economist Bronwyn Hughes Hall, who studies innovation and economics, are so frequently cited in other scholars' work _____ the usefulness of her research for her peers—other economists clearly find her studies valuable for their own scholarship.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) belies

(B) underscores

(C) forestalls

(D) overshadows

@JohannesKP
Highest scoring service
@JohannesKP

Shedding light on the thermal biology of fungi, research by Radamés Cordero et al. indicates that certain mushrooms (including *Boletus separans* and species from the genus *Thelephora*) can achieve a hypothermic state through evaporative cooling. Effects of this thermoregulation were not limited to the fungi's fruiting bodies and root-like hyphae: temperature reductions were observed in the air immediately surrounding the mushrooms. Though slight, the reductions inspired an air-cooling device; using approximately 400 grams of mushrooms, the team's prototype lowered the air temperature in a controlled environment by 10°C in forty minutes.

5 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

(A) It identifies an unexpected observation that motivated the study of evaporative cooling in fungi that is discussed earlier in the text.

(B) It provides empirical evidence to bolster the claim made earlier in the text that certain fungal species maintain a hypothermic state.

(C) It establishes a finding central to the text's discussion of a practical application resulting from the team's study of fungal thermal biology.

(D) It presents a tangential finding about thermoregulation in certain fungal species that the experiment described later in the text was designed to explain.

@JohannesKP Highest scoring service

In Grenada, use of solid fuel (e.g., coal, wood) as a share of total household fuel use fell by more than a third between 2000 and 2019; such shifts are often explained by appeal to the energy ladder, a model holding that fuel choice is mediated mainly by household income (specifically, high-technology fuels displace solid fuels as incomes rise). Moses Pundo and Gavin Fraser's study of fuel use in Kenya shows this model to be reductive, however: household fuel use was heterogeneous, flexible, and influenced by several factors, including the age of the household head.

6 Mark for Review

Which choice best describes the function of the information about Grenada in the text as a whole?

(A) It provides an example of a type of change that the text goes on to suggest is poorly suited for evaluating whether the energy ladder is a viable model.

(B) It introduces a finding that the text goes on to suggest can be explained in two different ways that are equally compelling.

(C) It illustrates the kind of phenomenon that the text goes on to suggest is frequently but inadequately accounted for by the energy ladder.

(D) It describes a trend that the text goes on to suggest has a similar cause as a seemingly unrelated trend observed in Kenya.

@JohannesKP

In a study by Mika R. Moran, Daniel A. Rodriguez, and colleagues, residents of Quito, Ecuador, and Lima, Peru, were surveyed about parks in their cities. Of the 618 respondents from Quito, 82.9% indicated that they use the city's parks, and of the 663 respondents from Lima, 72.7% indicated using city parks. Given that the percentage of Quito respondents who reported living within a 10-minute walk of a park was much lower than that reported by Lima respondents, greater proximity alone can't explain the difference in park use.

7 Mark for Review

The text makes which point about the difference between the proportions of Quito residents and Lima residents using parks?

(A) It was much larger than the researchers conducting the study expected.

(B) It could be due to inaccuracies in the survey results.

(C) It was calculated using sources that predate the survey.

(D) It is caused by something other than the parks' proximity to city residents.

@JohannesKP

The bird species *Schistocichla leucostigma* (the spot-winged antbird) shares some territory in French Guiana with *Thamnomanes caesius* (the cinereous antshrike), which emits a loud alarm call when it detects predators. Biologist Ari Martinez and colleagues recorded *T. caesius* alarm calls and played them in the vicinity of wild *S. leucostigma*. Finding that the birds often froze in place or scattered into vegetation upon hearing the calls, they concluded that *S. leucostigma* associates *T. caesius* alarm calls with danger.

8 Mark for Review

Which finding, if true, would most directly support Martinez and colleagues' conclusion?

(A) In some instances, *S. leucostigma* froze in place or scattered into vegetation when Martinez and colleagues approached but before they began playing sounds.

(B) When Martinez and colleagues played control sounds of random noise in the vicinity of *S. leucostigma*, the birds displayed no reaction.

(C) Other bird species than *S. leucostigma* also showed a tendency to freeze in place or scatter into vegetation when Martinez and colleagues played *T. caesius* alarm calls.

(D) Martinez and colleagues played alarm calls from different *T. caesius* individuals and observed no significant variation in the responses of *S. leucostigma*.

@JohannesKP

The Lego Group's introduction of the Legoland theme park in 1968 is a quintessential instance of brand extension—the company leveraged its brand recognition as a toy manufacturer to enter a product category where it had not previously competed. An outstanding question is whether perceived category similarity predicts consumers' likelihood of purchasing brand extensions. To answer this question, Alicia Grasby et al. identified 30 extended-brand pairs (e.g., the same brand of wristwatch and necktie) in 52 weeks of purchases by approximately 60,000 households and, for each pair, calculated the change in probability of a brand in one category being purchased if the same brand was purchased in the other category.

9 Mark for Review

Based on the text, which potential study design would be most likely to produce evidence that would enable Grasby et al. to answer their research question?

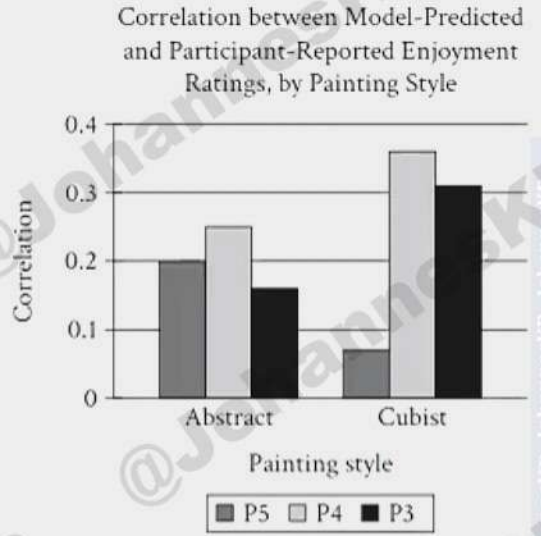
(A) Have a representative sample of the households rate the similarity of the product categories in each extended-brand pair, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.

(B) Poll a representative sample of the households to determine the degree of brand recognition for each brand in the extended-brand pairs, then determine how, if at all, the degree of brand recognition correlates with the frequency with which a different group of households purchased at least one product of that brand.

(C) Poll a representative sample of the households to determine the degree of brand recognition of each brand in the extended-brand pairs, then determine how, if at all, the degree of brand recognition correlates with the average cost of each product in the pairs.

(D) Have a representative sample of the households rate the similarity of one product in each extended-brand pair to other products in the same category, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.

@JohannesKP Highest scoring service



Neuroscientist Kiyohito Iigaya and colleagues developed a computational model to predict how much a person will enjoy a particular work of art on a scale from 1 (not at all) to 4 (very much). They then recruited participants to use the same scale to rate several sets of paintings in various styles and calculated the correlation between the ratings predicted by the model and those reported by the participants. Assuming participant P4 gave equal ratings to the abstract and cubist paintings, the data in the graph indicate the model predicted that _____

10 Mark for Review

Which choice most effectively uses data from the graph to complete the example?

(A) P4's ratings for abstract and cubist paintings would differ from one another.

(B) P4 would derive less aesthetic pleasure from cubist paintings than from abstract paintings.

(C) P4 would derive more aesthetic pleasure from cubist paintings than from abstract paintings.

(D) P4's rating for abstract and cubist paintings would equal one another.

@JohannesKP
Highest scoring service

Many studies have found a positive association between levels of dissolved organic carbon and mercury in bodies of fresh water in North America. But Petri Porvari and Matti Verta did not find this correlation in a study conducted in Finland, leading some scientists to hypothesize that the association is particular to North America. However, several other studies conducted outside North America, such as one by Sara M. Ekström and colleagues in Sweden, showed similar results to the North American studies, while few have produced results similar to those of Porvari and Verta's study, suggesting that

11 Mark for Review

Which choice most logically completes the text?

(A) the hypothesis that the positive association is particular to North America is correct.

(B) levels of dissolved organic carbon and mercury in bodies of fresh water are both much higher in Finland than elsewhere.

(C) there were circumstances unique to Ekström and colleagues' study that impeded accurate measurements of mercury levels.

(D) dissolved organic carbon and mercury levels do typically rise and fall together in fresh water.

Chelsea Wood et al. tracked temperature-driven changes in the abundance of *Anisakis* sp. (a complex life cycle parasite, or CLP, that requires three host species throughout its life cycle), *Gyrodactylus* sp. (a directly transmitted parasite, which requires only one host species), and 83 other parasite taxa found on eight fish species. CLPs are transmitted when an infected host is ingested by an individual of another species, typically shielding CLPs from the external environment, whereas directly transmitted parasites are exposed to external conditions during transmission. However, Wood et al. found that three-host CLP abundance decreased as sea temperatures rose, whereas directly transmitted parasite abundance was largely stable, suggesting that _____

12 Mark for Review

Which choice most logically completes the text?

(A) any advantages that the transmission strategy used by three-host CLPs may have conferred did not completely offset the negative effects of other temperature-driven factors on CLP abundance.

(B) CLPs primarily transmitted by ingestion were less dependent on host species adversely affected by warming temperatures than were CLPs that use other transmission strategies.

(C) as the number of host species involved in a parasite's transmission increases, the parasite is better protected against rising temperatures.

(D) directly transmitted parasites identified in the study were more likely to use transmission strategies that shield them from warming temperatures than were three-host CLPs.

@JohannesKP Highest scoring service

Prolonged exposure to anthropogenic noise (sounds from human sources like traffic or mining) can affect animals, as Amy Morris-Drake and colleagues found in a 2017 study of dwarf mongooses. Researchers conducted a meta-analysis of studies of how such noise affects animals and found that, for every study, relevant traits or behaviors of the animals were observably different between the exposed group and the otherwise similar but unexposed group. Although, on average, studies of mammals showed larger differences than studies of birds did, for every class of animals examined, there were individual studies showing differences well above the average for mammals. Therefore, the results of the meta-analysis suggest that _____

13 Mark for Review

Which choice most logically completes the text?

(A) the studies in the meta-analysis that examined mammals were more likely than those that examined birds to specify whether the observed effects were detrimental.

(B) the differences that studies attribute to exposure to anthropogenic noise are likely to be more pronounced for birds than they are for mammals.

(C) the difference found in the study conducted by Amy Morris-Drake and colleagues was likely larger than the average difference for studies of dwarf mongooses included in the meta-analysis.

(D) some studies of birds found larger effects of exposure to anthropogenic noise than some studies of mammals did.

@JohannesKP Highest Scoring service

Although the epic poem *The Aeneid* dates back to the 1st century BCE, _____ compelling narrative still captivates readers today.

14 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) their

(B) they're

(C) its

(D) it's

@JohannesKP
Highest scoring service

Most sand is beige because of deposits of gray- and tan-hued minerals, such as quartz and feldspar. The sand at Les Sables Roses Beach in French Polynesia is a more unusual _____ deposits of crushed coral and other organic matter lend the sand a unique pink hue.

15 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) shade, though;

(B) shade, though,

(C) shade, though

(D) shade; though

@JohannesKP
Highest Scoring Service

The poem "The Ballad of Rudolph Reed," which was published in 1960, contains three signature elements of Gwendolyn Brooks's poetic _____ compressed lines infused with bursts of vivid imagery; syncopated rhythms, largely inspired by the blues tradition; and a keen attention to everyday life in Brooks's South Side Chicago neighborhood.

16 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) style. Terse,

(B) style: terse,

(C) style, terse,

(D) style; terse,

@JohannesKP
Highest scoring service

Neoclassical economic models assume that people consistently make rational economic decisions, but Francesca Gino of Harvard Business School makes no such assumption; behavioral economists such as Gino, whose research focuses on worker productivity, _____ that economic decision-making can in fact be highly irrational.

17 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) contend

(B) is contending

(C) has contended

(D) contends

@JohannesKP
Highest scoring service

While the greater adjutant (*Leptoptilos dubius*) can be found in places like Dong Khanthung in Laos and Chitwan National Park in Nepal, more than 80 percent of this endangered stork species is found in Assam, India. There, wildlife _____ is on the front lines of conservation efforts to bring adjutants back from near extinction.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) biologist Dr. Purnima Devi Barman

(B) biologist: Dr. Purnima Devi Barman

(C) biologist, Dr. Purnima Devi Barman,

(D) biologist, Dr. Purnima Devi Barman

@JohannesKP
Highest Scoring service
@JohannesKP

The story of Chile's electrification is perhaps most readily understood by studying developments in several key years: 1916, when engineer Arturo Salazar attended the Pan-American Scientific Conference; 1936, when the study entitled "Política eléctrica chilena" was _____ 1943, when the national electrical plan was finalized.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) released; and

(B) released and

(C) released. And

(D) released, and

@JohannesKP
Highest scoring service

The Greek dramatist Euripides's *Cyclops*, a satyr play, is an extant work: it can still be read. By contrast, lost works like Euripides's play about the hero Alcmaeon, *Alcmaeon in Psophis*—no copy of which exists—_____ known to antiquarians only through references in extant works.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

(A) have become

(B) becomes

(C) has become

(D) is becoming

@JohannesKP Highest scoring service

Long thought to be sessile (immobile), adult *Chelonibia testudinaria*, barnacles that adhere to sea turtle shells, have been observed to shift slightly in position over time—a phenomenon that has been attributed to the barnacles' passive displacement by water currents. _____ a research team found that adult *C. testudinaria* moved toward the heads of their sea turtle hosts and thus against the prevailing water flow, behavior consistent with self-initiated locomotion.

21 Mark for Review

Which choice completes the text with the most logical transition?

(A) Confirming this hypothesis,

(B) Undermining this explanation,

(C) Drawing a similar conclusion,

(D) Contrary to this phenomenon,

Marcia Rieke is a space scientist who works on the James Webb Space Telescope, or JWST. Thanks in part to Rieke's contributions, the telescope is now positioned near the Sun-Earth L_2 Lagrange point, almost one million miles beyond Earth's orbit. _____ the JWST's predecessor, the Hubble Telescope, is only about 340 miles above Earth's surface.

22 Mark for Review

Which choice completes the text with the most logical transition?

(A) Therefore,

(B) Similarly,

(C) By contrast,

(D) Secondly,

@JohannesKP
Highest quality scoring service

Photogrammetry and polygonal modeling, methods used by digital artists to create three-dimensional elements for video games, typically require software to process geometric shapes numbering in the thousands. _____ these approaches are more economical than surface model methods, which require more powerful and expensive processors to manage polygons numbering in the millions.

23 Mark for Review

Which choice completes the text with the most logical transition?

(A) For example,

(B) Instead,

(C) As such,

(D) Specifically,

@JohannesKP
Highest Tutoring Service

The total solar eclipse of June 5, 762 BCE, was the second-longest total eclipse before the Dark Ages—7 minutes, 25 seconds long. Another memorable solar eclipse occurred on March 21, 424 BCE, but unlike the 762 BCE eclipse, the 424 BCE eclipse was annular. _____ the Moon didn't cover the Sun completely, instead creating an annulus, or "ring of fire."

24 Mark for Review

Which choice completes the text with the most logical transition?

(A) Meanwhile,

(B) That is,

(C) For example,

(D) Nonetheless,

@JohannesKP
Highest scoring service

While researching a topic, a student has taken the following notes:

- The Coachella Valley National Wildlife Refuge (NWR) is a protected natural area in California.
- It encompasses 3,592 acres.
- It was established to safeguard the habitat of the Coachella Valley fringe-toed lizard, an endangered species.
- The Coachella Valley NWR is managed by the US Fish & Wildlife Service.
- The US Fish & Wildlife Service limits human activities in the area.

25 Mark for Review

The student wants to indicate the size of the Coachella Valley NWR. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The Coachella Valley NWR is a natural area in California, home to the Coachella Valley fringe-toed lizard.

(B) A protected natural area, the Coachella Valley NWR encompasses 3,592 acres of land in California.

(C) The Coachella Valley NWR is a protected natural area managed by the US Fish & Wildlife Service, which limits human activities there.

(D) Home to the Coachella Valley fringe-toed lizard, California's Coachella Valley NWR is managed by the US Fish & Wildlife Service.

@JohannesKP Highest scoring service

While researching a topic, a student has taken the following notes:

- Antelope House is an Ancestral Puebloan dwelling site located in northeastern Arizona.
- It was built under a rock overhang and inhabited from approximately 1050–1270 CE.
- The overhanging rock ledges offered protection from heavy rain and snow.
- Mud Springs Pueblo is an Ancestral Puebloan dwelling site located in southwestern Colorado.
- It was built on relatively flat terrain and inhabited from approximately 1200–1250 CE.
- This level surface allowed for the construction of large terraced buildings.

26 Mark for Review

The student wants to explain an advantage of the Antelope House dwelling site. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) The relatively flat terrain on which Antelope House was built allowed for the construction of large terraced buildings.

(B) Located in northeastern Arizona, Antelope House is an Ancestral Puebloan dwelling site that was inhabited from approximately 1050–1270 CE.

(C) The location of Antelope House, an Ancestral Puebloan dwelling site in northeastern Arizona, provided an advantage to its inhabitants.

(D) Since it was built under a rock overhang, Antelope House was naturally protected from heavy rain and snow.

@JohannesKP Highest Scoring service

While researching a topic, a student has taken the following notes:

- Kepler's first law of planetary motion states that the orbit of a planet around the Sun is an ellipse.
- The law also states that, in an elliptical orbit, the object being orbited is one of the ellipse's foci.
- Kepler's laws of planetary motion also apply to natural satellites (e.g., moons).
- Europa is a moon of Jupiter that orbits the planet in 3.55 Earth days on average.
- Europa's orbit is elliptical.

27 Mark for Review

The student wants to provide an explanation and example of Kepler's first law of planetary motion. Which choice most effectively uses relevant information from the notes to accomplish this goal?

(A) Kepler's first law of planetary motion, which describes the orbits of Jupiter and other planets in the solar system, states that the object being orbited is one of the ellipse's foci.

(B) Jupiter's moon Europa completes an orbit in 3.55 Earth days on average, a clear example of Kepler's first law of planetary motion, which describes the elliptical orbit of planets.

(C) Kepler's first law of planetary motion states that the orbit of a planet around the Sun is an ellipse; for example, planetary satellites orbit their planets in an elliptical fashion.

(D) Europa's orbit of Jupiter is elliptical, demonstrating Kepler's first law of planetary motion, which describes the elliptical orbits of planets but applies to moons as well.

@JohannesKP Highest scoring service

1 Mark for Review

The function g is defined by $g(x) = -18 - 9x$. For what value of x does $g(x) = -216$?

(A) -26

(B) 22

(C) 26

(D) $1{,}926$

@JohannesKP
Highest scoring service

2 Mark for Review

Which expression is equivalent to $13(x^2 - 7)$?

(A) $13x^2 - 91$

(B) $13x^2 - 20$

(C) $13x^2 - 7$

(D) $13x^2 + 6$

@JohannesKP
Highest scoring service

3 Mark for Review

The function $f(x) = 600(0.5)^{\frac{x}{11}}$ gives the predicted intensity of a beam $f(x)$, in number of photons in the beam, x millimeters below the surface of a certain material. What is the best interpretation of $f(11) = 300$?

(A) A beam at the surface of the material has a predicted intensity of 11 photons in the beam.

(B) A beam at the surface of the material has a predicted intensity of 300 photons in the beam.

(C) A beam 11 millimeters below the surface of the material has a predicted intensity of 300 photons in the beam.

(D) A beam 300 millimeters below the surface of the material has a predicted intensity of 600 photons in the beam.

@JohannesKP

4 Mark for Review

A sleep study consisted of 59 participants, of which 53 participants each had an average of more than 120 minutes of rapid eye movement (REM) sleep per night. If a participant from this sleep study is selected at random, what is the probability of selecting a participant that had an average of more than 120 minutes of REM sleep per night?

(A) $\frac{53}{100}$

(B) $\frac{59}{100}$

(C) $\frac{53}{59}$

(D) $\frac{59}{53}$

Highest scoring service
@JohannesKP

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
$-\frac{1}{3}$	-1/3 -.3333	-.33

5 Mark for Review

For certain air temperatures, the table gives wind chill temperatures for two different wind speeds.

Air temperature	Wind chill temperature at wind speed 15 mph	Wind chill temperature at wind speed 45 mph
26°F	14°F	7°F
30°F	19°F	12°F
34°F	24°F	18°F

According to the table, what is the wind chill temperature, in degrees Fahrenheit (°F), when the air temperature is **26°F** and the wind speed is **15 miles per hour (mph)**? (Disregard the degree symbol when entering your answer.)

14

Answer Preview: 14

Show Keypad

@JohannesKP
Highest scoring service

6 Mark for Review

A soccer team's goal is to earn at least $1,320 by selling coupon books. The team earns $12 from selling each coupon book. Which of the following inequalities describes all possible values for the number of coupon books, n, the team can sell to meet the goal?

(A) $n + 12 \leq 1{,}320$

(B) $n + 12 \geq 1{,}320$

(C) $12n \leq 1{,}320$

(D) $12n \geq 1{,}320$

@JohannesKP
Highest scoring service

7 Mark for Review

$$2.5c + 5d = 60$$

The given equation describes the relationship between the number of cats, c, and the number of dogs, d, that can be cared for at a pet care business during a week. If the business cares for 12 dogs during a week, how many cats can it care for during this week?

(A) 0

(B) 5

(C) 30

(D) 60

@JohannesKP
Highest scoring service

8 Mark for Review

$$-(3x-7)^2 = p + 17$$

In the given equation, p is an integer constant. The equation has no real solution. What is the least possible value of p?

(A) -18

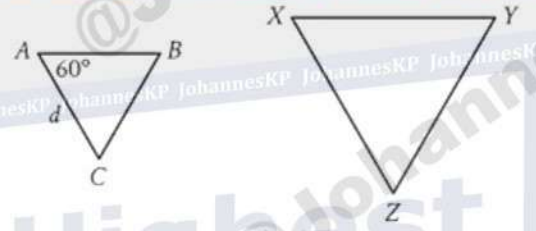
(B) -17

(C) -16

(D) -15

@JohannesKP Highest scoring service

9 Mark for Review



Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 5 to obtain triangle XYZ, where $d = 19$. What is the measure, in degrees, of angle X?

(A) 12

(B) 55

(C) 60

(D) 65

10 Mark for Review

The circumference of the base of a right circular cylinder is 6π meters, and the height of the cylinder is 19 meters. What is the volume, in cubic meters, of the cylinder?

(A) 57π

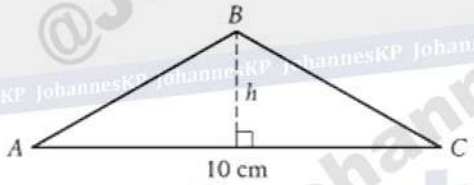
(B) 114π

(C) 171π

(D) 684π

@JohannesKP Highest scoring service

11 Mark for Review



Note: Figure not drawn to scale.

The area of triangle ABC is 220 square centimeters. What is the height h, in centimeters, of this triangle?

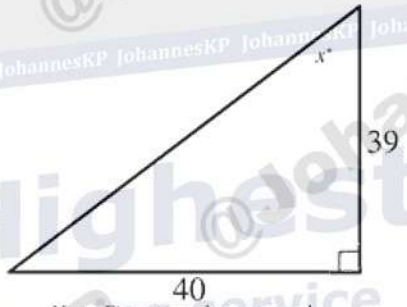
(A) 10

(B) 22

(C) 44

(D) 88

@JohannesKP
Highest scoring service



Note: Figure not drawn to scale.

In the right triangle shown, the value of $\tan x^\circ$ is $\frac{c}{39}$. What is the value of c?

(A) 79

(B) 40

(C) 39

(D) 1

@JohannesKP

13 Mark for Review

The graph of the polynomial function f in the xy-plane, where $y = f(x)$, has x-intercepts of $(-5, 0)$ and $(6, 0)$. Which of the following must be true?

(A) $f(0) = -5$

(B) $f(6) = -5$

(C) $f(-5) = 0$

(D) $f(-5) = 6$

@JohannesKP Highest scoring service

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

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3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
1	-1/3	-.33

14 Mark for Review

$$(x-16)(x-10)(x+7)(x+17)=0$$

What is a positive solution to the given equation?

10

Answer Preview: 10

Show Keypad

Highest scoring service
@JohannesKP

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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1	-1/3 -.3333 -.3334	-.33

16 Mark for Review

In the xy-plane, the graph of the equation $(x-6)^2+(y-2)^2=36$ is a circle. The point $(12, c)$, where c is a constant, lies on this circle. What is the value of c?

Answer Preview:

Show Keypad

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
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	-1/3	

17 Mark for Review

$$6(x-4) = (x-4) + 6$$

If x is the solution to the given equation, what is the value of $x-4$?

6/5

Answer Preview: $\frac{6}{5}$

Show Keypad

@JohannesKP Highest scoring service

18 Mark for Review

$$\frac{5}{4}x + 2y = 17$$

$$\frac{3}{4}x + 2y = 15$$

The solution to the given system of equations is (x, y). What is the value of $\frac{11}{4}x + 6y$?

(A) 4

(B) 6

(C) 32

(D) 47

@JohannesKP Highest scoring service

19 Mark for Review

The function h is defined by $h(x) = 14(2)^{-x} + 7$. What is the value of $h(3)$?

(A) $\frac{42}{4}$

(B) $\frac{35}{4}$

(C) -77

(D) -105

@JohannesKP Highest scoring service

Student-produced response directions

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- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
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	-1/3	

20 Mark for Review

The graph of a line in the xy-plane passes through the point $(1, 8)$ and crosses the x-axis at the point $(7, 0)$. The line crosses the y-axis at the point $(0, b)$. What is the value of b?

28

Answer Preview: $\frac{28}{1}$

Show Keypad

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
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1	-1/3	-.33

x	y
-2	19
0	31
2	43

The table shows three values of x and their corresponding values of y. The linear relationship between x and y can be represented by an equation written in the form $Ax + By = C$, where A, B, and C are constants. What is the value of $\frac{A}{B}$?

Answer Preview:

Show Keypad

@JohannesKP Highest scoring service

$$5x + 8y = 9$$
$$15x + 24y = 27$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

(A) $(r, -\frac{5r}{8} + \frac{9}{8})$

(B) $(-\frac{5r}{8} + \frac{9}{8}, r)$

(C) $(-\frac{5r}{8} + 9, \frac{5r}{8} + 27)$

(D) $(\frac{r}{3} + 9, -\frac{r}{3} + 27)$

@JohannesKP Highest scoring service

1 Mark for Review

Julia purchased 800 feet of fencing. She used 60% of this fencing to surround a vegetable garden. How many feet of fencing did Julia use to surround the vegetable garden?

(A) 12

(B) 30

(C) 480

(D) 740

Highest scoring service @JohannesKP

2 Mark for Review

The population of a certain city doubled every 25 years from 1660 to 1760. The population of this city was 208,000 in 1760. What was the population of this city in 1660?

(A) 13,000

(B) 26,000

(C) 104,000

(D) 416,000

@JohannesKP Highest scoring service

3 Mark for Review

1, 4, 7, 10, 33

What is the mean of the data shown?

(A) 7

(B) 11

(C) 17

(D) 33

@JohannesKP
Highest scoring service

4 Mark for Review

For the linear function h, $h(0) = 0$ and $h(11) = 5$. Which equation defines h?

(A) $h(x) = \frac{5}{11}x$

(B) $h(x) = \frac{5}{11}x + 5$

(C) $h(x) = \frac{11}{5}x$

(D) $h(x) = \frac{11}{5}x + 11$

@JohannesKP Highest scoring service

5 Mark for Review

Hassan put up wire fencing along each edge of a rectangular garden with a length of x feet and a width of y feet. Hassan put up a total length of 44 feet of wire fencing. Which equation represents this situation?

(A) $2x + 2y = 44$

(B) $x + y = 44$

(C) $2xy = 44$

(D) $xy = 44$

@JohannesKP

6 Mark for Review

How many solutions does the equation $7x - 19 = 19 - 7x$ have?

(A) Zero

(B) Exactly one

(C) Exactly two

(D) Infinitely many

@JohannesKP
Highest scoring service

7 Mark for Review

$$y > 17$$

$$4x + y < 21$$

The point $(x, 54)$ is a solution to the given system of inequalities in the xy-plane. Which of the following could be the value of x?

(A) 9

(B) 5

(C) -5

(D) -9

@JohannesKP
Highest scoring service

8 Mark for Review

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is $\$190$ when it sells 4 items, and its profit is $\$670$ when it sells 10 items. Which equation defines f?

(A) $f(x) = 180x - 670$

(B) $f(x) = 67x$

(C) $f(x) = 80x - 10x$

(D) $f(x) = 80x - 130$

@JohannesKP Highest scoring service

9 Mark for Review

Which quadratic equation has no real solutions?

(A) $-2x^2 = 0$

(B) $-2x^2 + 2 = 0$

(C) $-2x^2 + 2x = 0$

(D) $-2x^2 + 2x - 2 = 0$

@JohannesKP Highest scoring service

10 Mark for Review

What is the y-intercept of the graph of $y = 16^{x+3}$ in the xy-plane?

A $(0, 4{,}096)$

B $(0, 0)$

C $(0, 16)$

D $(0, 3)$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67
1	-1/3 -.3333	-.33

12 Mark for Review

A carnival receives $10 for each adult admission ticket sold and $5 for each child admission ticket sold. On a particular day, the carnival received a total of $4,100 from selling adult and child admission tickets. If 220 adult admission tickets were sold on that day, how many child admission tickets were sold?

380

Answer Preview: 380

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13 Mark for Review

$$y - x = 32$$

$$y = x^2 - 30x$$

The graphs of the equations in the given system intersect at the point (x, y) in the xy-plane. What is a possible value of y?

(A) 32

(B) 33

(C) 63

(D) 64

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14 Mark for Review

$$c(r) = \frac{2}{5}(2\pi r)$$

The function c gives the partial length, in centimeters (cm), of the circumference of a circle with a radius that measures r cm. According to the function, for each increase of 18 cm in the length of the radius, this partial length of the circumference increases by $k\pi$ cm. What is the value of k?

72/5

Answer Preview: $\frac{72}{5}$

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15 Mark for Review

$$f(x) = 25(1.40)^{\frac{x}{6}}$$

For the given function f, the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p?

(A) 40

(B) 49

(C) 80

(D) 96

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16 Mark for Review

The function k is defined by $k(s) = \sqrt{s + 110}$. If $k(53p) = p$, where p is a constant, what is the value of p?

55

Answer Preview: 55

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17 Mark for Review

If $121x^2 = -110x + 56$, what is the value of $11x + 5$, where $11x + 5 > 0$?

9

Answer Preview: 9

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18 Mark for Review

A computer program models the population of a certain insect in an environment where the insect has no natural predators. According to the model, the estimated total mass of the population of insects at the end of every 5-week period is 158% greater than the estimated total mass of the population of insects at the end of the previous 5-week period. The estimated total mass of the population of insects at the end of 15 weeks is 627.7 grams. Which equation best represents this model, where M is the estimated total mass, in grams, of the population of insects at the end of t weeks?

(A) $M = 538.92(1.58)^{\frac{t}{5}}$

(B) $M = 457.66(2.58)^{\frac{t}{5}}$

(C) $M = 159.14(1.58)^{\frac{t}{5}}$

(D) $M = 36.55(2.58)^{\frac{t}{5}}$

19 Mark for Review

The value of an autographed baseball increased by 178% from the end of 2012 to the end of 2013 and then decreased by 21% from the end of 2013 to the end of 2014. What was the net percentage increase in the value of the autographed baseball from the end of 2012 to the end of 2014?

(A) 119.62%

(B) 140.62%

(C) 157.00%

(D) 236.38%

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20 Mark for Review

If n and k are numbers greater than 1 and $\sqrt[4]{n^5}$ is equivalent to $\sqrt[3]{k^2}$, for what value of a is n^{2a+1} equal to k?

7/16

Answer Preview: $\frac{7}{16}$

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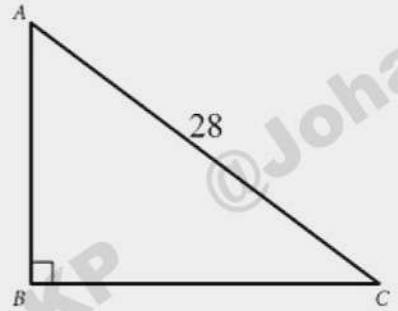
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21 Mark for Review



Note: Figure not drawn to scale.

For triangle ABC, the length of $\overline{AB}$ is 11 less than the length of $\overline{AC}$. Point D (not shown) lies on $\overline{AC}$ such that $\overline{BD}$ (not shown) is perpendicular to $\overline{AC}$. What is the value of $\frac{BC}{BD}$?

28

Answer Preview: 28

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22 Mark for Review

The area of a rectangular region is increasing at a rate of 280 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

(A) 0.43

(B) 1.42

(C) 15.31

(D) 26.03

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